

the share is mounted, copy the proposal document that you are working on from your machine to this share, and we are then ready to set up snapshots and their management strategy.

You can refer to the screenshots in Figures 6 through Figure 10 for clarity.

Setting up snapshots

For setting automatic snapshots, go to 'Scheduled Tasks' under 'System' (and click 'Schedule a Task') or 'Snapshots' under 'Storage' (and click 'Schedule') of the Web UI (basically, these are two ways of getting to the same place). The images are shown in Figures 11 and 12. You can configure Rockstor to take snapshots every 30 minutes, an hour, day, week, month, year, or across multiple time periods. This way, you can recover whatever copy you had saved at a particular time.

As Snapshots are created per schedule, these will start appearing inside the share (Marketing_Proposals). Going back to our example above, if you want to retrieve the paragraph you edited and removed, and compare it to your current work, you can simply retrieve your back-up from the last 30 minutes or past hour. Figure 12 shows how snapshots are visible on your mounted folder. If you click inside a folder, you will have prior versions of your snapshot. Snapshots, therefore, ensure that you not just have the current version of your work but also a trail of all prior versions saved over a period of time. Figure 11 shows how the set-up screen for a snapshot looks.

Manage your backups using snapshots

Backups can be managed by using multi-period snapshots, a technique or strategy that allows you to schedule snapshots across multiple time periods. If you are actively working on a project for a week, you may want to back up every 30 minutes or hour of your work while you are actively working on it during that week. However, once you are done, you may want to back up every month, and then every year for archival purposes. Multi-period snapshots help you achieve this 'self-thinning' and manage your backups and storage space.

I will now describe one multi-period snapshot and self-thinning backup management strategy for the proposal document. You can modify this strategy or come up with one that fits your needs better.

While setting up a snapshot task, there is a field called *Maximum count of snapshots* where you can specify the maximum number of snapshots you want (Figure 11). Say you are setting up an hourly backup, and you want to back up every hour for seven days of a week, then you can set the maximum count to 250 (greater than 24×7) snapshots. Once the snapshot count reaches 250, the newer snapshots start overwriting



Figure 13: Self-thinning backups representation



Figure 14: Multi-period snapshots carried over time

the older ones. Let us also assume that your project was a week long, and for now, you are done, but may make minor edits later. So, you want to retain the 168th snapshot, carry it forward and delete the rest. Say, you set up the snapshot tasks at 10 p.m. on a Monday, so the 168th snapshot will be taken at 9 p.m. on Sunday. You can set up a weekly snapshot task for 9 p.m. of Sunday of that week and move off the daily backups.

You could also choose to carry forward this weekly snapshot for a month. And you can choose a snapshot for a particular day (say, 9 p.m. on the third Sunday of September) and delete the rest. So, now your snapshots will be taken every month. This allows you to make minor changes to your backup and then save a copy every month.

Assuming that you have no need for a monthly backup, and now you want to retain one snapshot for a month in a year for archival purposes and delete the rest, you can do that too (See Figures 13 and 14).

This way, you can start with a lot of snapshots (starting with 168 a week) when you are actively working on a project, and then just save one copy of the final version for archiving.

Similarly, you could create your own strategy for daily, weekly, monthly and yearly snapshots and backups. **END** 

References

- [1] <http://rockstor.com>.
- [2] <http://rockstor.com/docs/>
- [3] <http://forum.rockstor.com/>

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